MESSAGE INTELLIGENCE SYSTEM

PART A — PROBABILITY & CONCEPTUAL FOUNDATION

Theory Notes | Supervised Learning

1. What is Conditional Probability?

Conditional probability is the probability of an event occurring when another event is already known to have occurred. It is written as P(A|B), meaning the probability of event A given event B, where P(B) is greater than zero. For message classification, it can describe the probability that a message is spam given that it contains a suspicious link.

Formula: $P(A|B) = P(A \cap B) / P(B)$

2. Explain Bayes' Theorem and its importance in classification problems.

Bayes' Theorem updates the probability of a class using observed evidence. It combines the prior probability of a class with the likelihood of the observed features to calculate a posterior probability. In spam detection, it helps estimate the probability that a message is spam given features such as keywords, URLs, or message length.

Formula: $P(A|B) = [P(B|A) \times P(A)] / P(B)$

3. What assumptions does the Naive Bayes Classifier make?

Naive Bayes assumes that the input features are conditionally independent of one another once the class label is known. For example, given that a message is spam, the model treats features such as URL count and keyword frequency as independent for the purpose of calculating class probabilities. This assumption simplifies computation, although real features may be correlated.

4. Explain the working principle of KNN and SVM.

4(a). K-Nearest Neighbors (KNN)

KNN is a distance-based classifier that stores the training examples and classifies a new example by finding its K closest neighbors in feature space. The predicted class is usually the majority class among those neighbors. Feature scaling is important because variables with larger numeric ranges can otherwise dominate distance calculations.

4(b). Support Vector Machine (SVM)

SVM is a margin-based classifier that finds a decision boundary (hyperplane) separating classes while maximizing the margin between them. The closest training points are called support vectors. Linear kernels model a straight boundary, while kernels such as RBF can represent more complex, non-linear boundaries.

5. Compare distance-based, probabilistic, and margin-based classifiers.

Distance-based classifiers such as KNN use similarity or distance between examples. Probabilistic classifiers such as Naive Bayes estimate class probabilities using probability rules and model assumptions. Margin-based classifiers such as SVM choose a decision boundary that separates classes with a large margin. Their performance depends on the data, feature representation, and model settings.

Application to the Project

In the Message Intelligence System, these concepts support the comparison of KNN, SVM, and Naive Bayes for classifying messages as legitimate (0) or spam (1). The models can be evaluated using accuracy, precision, recall, and F1 score, while remembering that no single metric fully describes model performance.

Prepared from Part A of the supplied project brief.